# Problem Set 3

Zahra Khan

Due: November 13, 2025

## Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

## Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```r
#running the linear regression
regression_model_problem1 <- lm(voteshare ~ difflog , data=inc.sub)
summary(regression_model_problem1)
```

Several key observations are highlighted in the regression table below. First, a one-unit increase in difflog is associated with a 0.042 increase in voteshare. The value of voteshare when difflog equals zero is 0.579. Additionally, the R-squared value is 0.367, indicating that approximately 36.7% of the variation in Y can be attributed to the model.

Table 1:

| | *Dependent variable:* |
|---|---|
| | voteshare |
| difflog | 0.042*** |
| | (0.001) |
| Constant | 0.579*** |
| | (0.002) |
| Observations | 3,193 |
| $R^2$ | 0.367 |
| Adjusted $R^2$ | 0.367 |
| Residual Std. Error | 0.079 (df = 3191) |
| F Statistic | 1,852.791*** (df = 1; 3191) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

2. Make a scatterplot of the two variables and add the regression line.

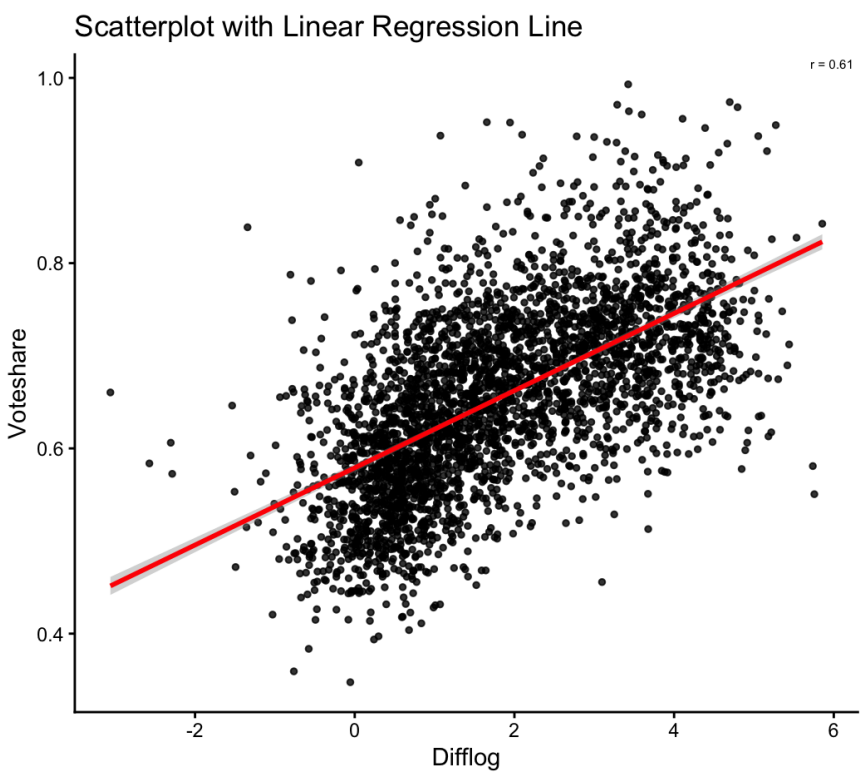


Figure 1: Model from Question 1

```r
#saving correlation coefficient for plot
m1_corr <- cor.test(inc.sub$difflog, inc.sub$voteshare, method = "pearson")
r1 <- m1_corr$estimate
#Creating Scatterplot
ggplot(inc.sub, aes(x = difflog, y = voteshare)) +
  geom_point(color = "black", size = 1, alpha = 0.8) +     # Scatter points
  geom_smooth(method = "lm", color = "red", se = TRUE, linewidth = 1) +
  annotate(
    "text",
    x = Inf, y = Inf,                                       # place in top-right corner
    label = paste0("r = ", round(r1, 2)),          # correlation label
    hjust = 1.1, vjust = 1.5, size = 2, color = "black"
  ) +
  theme_classic(base_size = 11) +
  labs(
    title = "Scatterplot with Linear Regression Line",
    x = "Difflog",
    y = "Voteshare"
  )
```

3. Save the residuals of the model in a separate object.

```r
res1_model <- regression_model_problem1$residuals
```

4. Write the prediction equation.

$$\hat{y} = b_0 + b_1 x$$

$$\widehat{\text{voteshare}} = 0.579 + 0.042 \times \text{difflog}$$

# Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

```
regression_model_problem2 <- lm(presvote ~ difflog, data=inc.sub)
summary(regression_model_problem2)
```

Several key observations are worth highlighting from the regression. First, a one-unit increase in difflog(beta) is associated with a 0.024 unit increase in presvote. Additionally, the value of presvote when difflog is zero is 0.508. Finally, the R-squared value is 0.088, which means that about 8.8% of the variation in presvote can be explained by the model.

Table 2:

| | *Dependent variable:* |
|---|---|
| | presvote |
| difflog | 0.024*** |
| | (0.001) |
| Constant | 0.508*** |
| | (0.003) |
| Observations | 3,193 |
| $R^2$ | 0.088 |
| Adjusted $R^2$ | 0.088 |
| Residual Std. Error | 0.110 (df = 3191) |
| F Statistic | 307.715*** (df = 1; 3191) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

2. Make a scatterplot of the two variables and add the regression line.

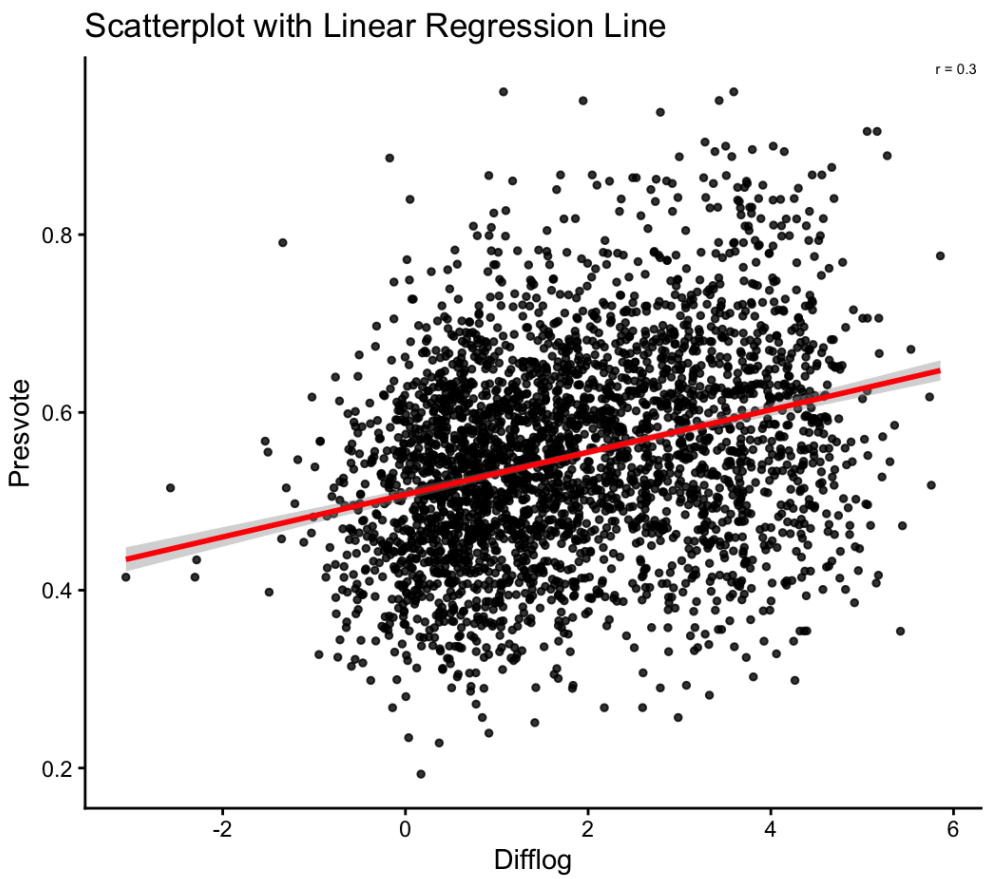


Figure 2: Model from Question 2

```
#Creating Scatterplot
ggplot(inc.sub, aes(x = difflog, y = presvote)) +
  geom_point(color = "black", size = 1, alpha = 0.8) +     # Scatter points
  geom_smooth(method = "lm", color = "red", se = TRUE, linewidth = 1) +
  annotate(
    "text",
    x = Inf, y = Inf,                                      # place in top-right corner
    label = paste0("r = ", round(r2, 2)),          # correlation label
    hjust = 1.1, vjust = 1.5, size = 2, color = "black"
  ) +
  theme_classic(base_size = 11) +
  labs(
    title = "Scatterplot with Linear Regression Line",
    x = "Difflog",
    y = "Presvote"
  )
```

3. Save the residuals of the model in a separate object.

```
res2_model <- regression_model_problem2$residuals
```

4. Write the prediction equation.

$$\hat{y} = b_0 + b_1 x$$

$$\widehat{\text{presvote}} = 0.508 + 0.024 \times \text{difflog}$$

# Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

```
regression_model_problem3 <- lm(voteshare ~ presvote, data=inc.sub)
summary(regression_model_problem3)
```

There a a few key observations to highlight from the regression. First, a one-unit increase in presvote (beta) is associated with a 0.388 increase in voteshare. The value of voteshare when presvote is zero is 0.441. In addition to this, the R-squared value is 0.206, which means approximately 0.21% of the variation in voteshare is explained by the model.

Table 3:

| | *Dependent variable:* |
|---|---|
| | voteshare |
| presvote | 0.388*** |
| | (0.013) |
| Constant | 0.441*** |
| | (0.008) |
| Observations | 3,193 |
| $R^2$ | 0.206 |
| Adjusted $R^2$ | 0.206 |
| Residual Std. Error | 0.088 (df = 3191) |
| F Statistic | 826.950*** (df = 1; 3191) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

2. Make a scatterplot of the two variables and add the regression line.

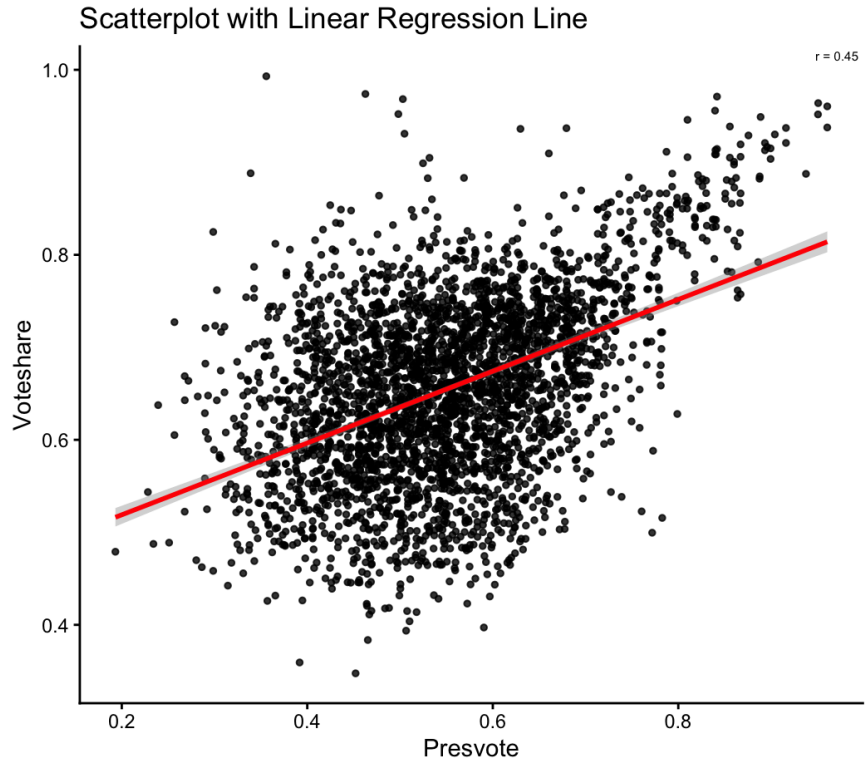


Figure 3: Model from Question 3

```
#Saving Correlation Coefficient
m3_corr <- cor.test(inc.sub$presvote, inc.sub$voteshare, method = "pearson")
r3 <- m3_corr$estimate
#Creating Scatterplot
ggplot(inc.sub, aes(x = presvote, y = voteshare)) +
  geom_point(color = "black", size = 1, alpha = 0.8) +     # Scatter points
  geom_smooth(method = "lm", color = "red", se = TRUE, linewidth = 1) +
  annotate(
    "text",
    x = Inf, y = Inf,                                      # place in top-right corner
    label = paste0("r = ", round(r3, 2)),          # correlation label
    hjust = 1.1, vjust = 1.5, size = 2, color = "black"
  ) +
  theme_classic(base_size = 11) +
  labs(
    title = "Scatterplot with Linear Regression Line",
    x = "Presvote",
    y = "Voteshare"
  )
```

3. Write the prediction equation.

$$\hat{y} = b_0 + b_1 x$$

$$\widehat{\text{voteshare}} = 0.441 + 0.39 \times \text{presvote}$$

# Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```r
regression_model_problem4 <- lm(res1_model ~ res2_model, data = inc.sub)
summary(regression_model_problem4)
```

There are a few key points to highlight regarding the regression. A one-unit increase in residual 2 is associated with a 0.257 increase in residual 1. Secondly, the R-squared value is 0.130, which means about 13% of the variation in residual 1 can be explained by the model.

Table 4:

| | *Dependent variable:* |
|---|---|
| | res1_model |
| res2_model | 0.257*** |
| | (0.012) |
| Constant | −0.000 |
| | (0.001) |
| Observations | 3,193 |
| $R^2$ | 0.130 |
| Adjusted $R^2$ | 0.130 |
| Residual Std. Error | 0.073 (df = 3191) |
| F Statistic | 476.975*** (df = 1; 3191) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

2. Make a scatterplot of the two residuals and add the regression line.

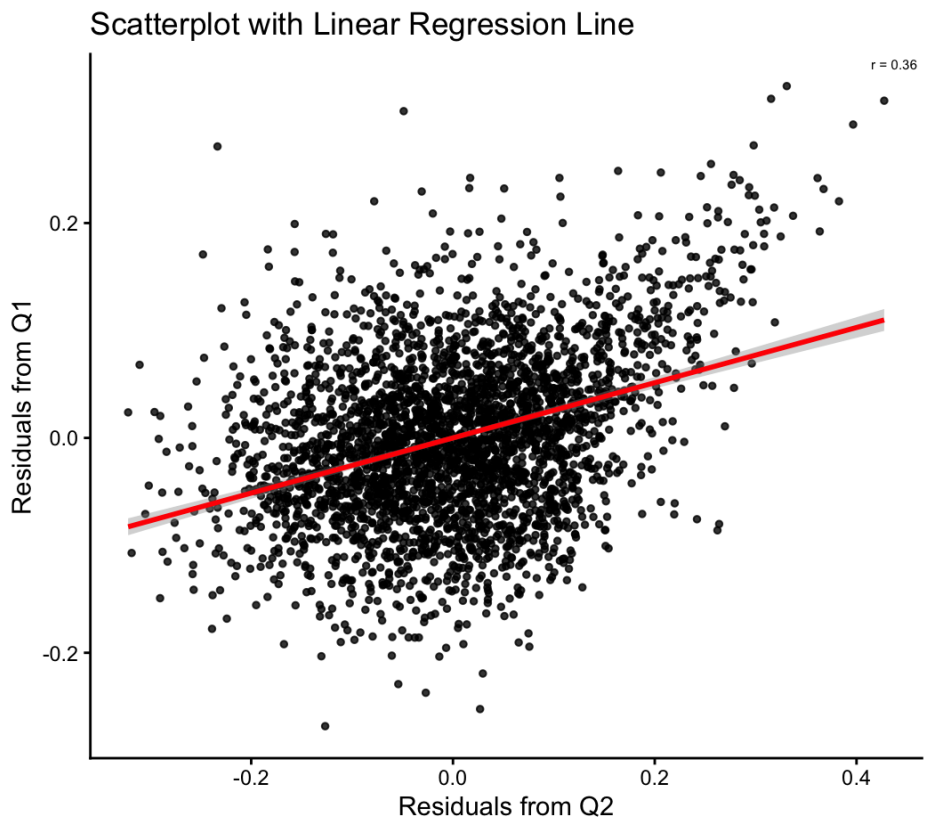


Figure 4: Model from Question 4

```
#Saving Correlation Coefficient
m4_corr <- cor.test(res1_model, res2_model, method = "pearson")
r4 <- m4_corr$estimate
#Creating Scatterplot
ggplot(inc.sub, aes(x = res2_model, y = res1_model)) +
  geom_point(color = "black", size = 1, alpha = 0.8) +     # Scatter points
  geom_smooth(method = "lm", color = "red", se = TRUE, linewidth = 1) +
  annotate(
    "text",
    x = Inf, y = Inf,                                      # place in top-right corner
    label = paste0("r = ", round(r4, 2)),                  # correlation label
    hjust = 1.1, vjust = 1.5, size = 2, color = "black"
  ) +
  theme_classic(base_size = 11) +
  labs(
    title = "Scatterplot with Linear Regression Line",
    x = "Residuals from Q2",
    y = "Residuals from Q1"
  )
```

3. Write the prediction equation.

$$\hat{y} = b_0 + b_1 x$$

$$\widehat{\text{residualone}} = 0.00 + 0.257 \times \text{residualtwo}$$

# Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```r
regression_model_problem5 <- lm(voteshare ~ difflog + presvote, data =
    inc.sub)
summary(regression_model_problem5)
```

Here are a few crucial observations to state about the regression. First, on average, a one-unit change in difflog is associated with a 0.036-unit change in voteshare, holding all other variables constant. In addition to this, on average, a one-unit increase in preference is associated with a 0.257-unit increase in vote share, holding everything else constant. The R-squared value has increased from previous models, and now 45% of the variation in voteshare can be explained by the model.

Table 5:

| | *Dependent variable:* |
|---|---|
| | voteshare |
| difflog | 0.036*** |
| | (0.001) |
| presvote | 0.257*** |
| | (0.012) |
| Constant | 0.449*** |
| | (0.006) |
| Observations | 3,193 |
| $R^2$ | 0.450 |
| Adjusted $R^2$ | 0.449 |
| Residual Std. Error | 0.073 (df = 3190) |
| F Statistic | 1,302.947*** (df = 2; 3190) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

2. Write the prediction equation.

$$Y = b_0 + b_1X_1 + b_2X_2 + \cdots + b_nX_n + \varepsilon$$

$$\widehat{\text{voteshare}} = 0.449 + 0.036 \times \text{difflog} + 0.257 \times \text{presvote}$$

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

Table 6:

| | *Dependent variable:* | |
|---|---|---|
| | res1_model | voteshare |
| | (1) | (2) |
| res2_model | 0.257*** | |
| | (0.012) | |
| difflog | | 0.036*** |
| | | (0.001) |
| presvote | | 0.257*** |
| | | (0.012) |
| Constant | −0.000 | 0.449*** |
| | (0.001) | (0.006) |
| Observations | 3,193 | 3,193 |
| $R^2$ | 0.130 | 0.450 |
| Adjusted $R^2$ | 0.130 | 0.449 |
| Residual Std. Error | 0.073 (df = 3191) | 0.073 (df = 3190) |
| F Statistic | 476.975*** (df = 1; 3191) | 1,302.947*** (df = 2; 3190) |
| *Note:* | | *p<0.1; **p<0.05; ***p<0.01 |

When comparing the two regression outputs, we see that the residuals from model two are equal to the estimated coefficient value of the presvote variable. This is the case, given that the variation in voteshare attributed to presvote remains unaffected by adding difflog.